Rational curves in the Kähler MMP and a Kähler criterion

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Kähler varieties

Definition (Kähler varieties)

Let X be a normal complex variety. A Kähler form on X is a closed real $(1,1)$-form ω which is locally of the form $i\partial\bar{\partial}\varphi$ with φ smooth and strictly plurisubharmonic. A Kähler variety is a complex variety with a Kähler form.

Definition (Fujiki varieties)

A compact complex variety X is in Fujiki's class $\mathcal{C}$ if it is bimeromorphic to a compact Kähler manifold.

Examples include compact Kähler varieties and *Moishezon* varieties, i.e. varieties bimeromorphic to a projective one; Hironaka's example below is of this kind.

Kähler surfaces

Theorem (Kodaira, Miyaoka, Siu, Lamari, Buchdahl)

A compact complex surface is Kähler if and only if b_1 is even.

Theorem (Takayama [Tak03])

If X has klt singularities (e.g. X smooth) and $\widetilde{X} \to X$ is a resolution, then $\pi_1(\widetilde{X}) \cong \pi_1(X)$. In particular π_1, and hence b_1, is a bimeromorphic invariant.

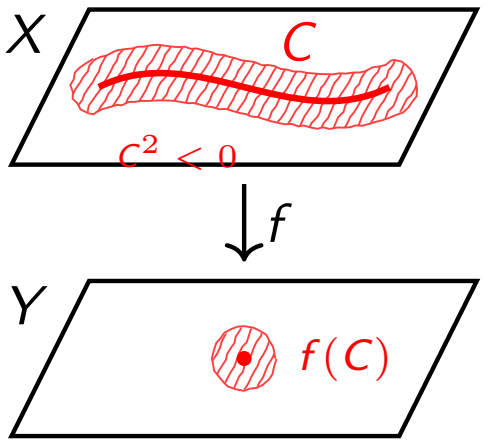
Hence being Kähler is a bimeromorphic invariant for complex surfaces.

Projective surfaces

Let X be a smooth projective surface and $C \subset X$ a curve with $C^2 < 0$. Since $D \cdot C \geq 0$ for every curve $D \neq C$, the class of C spans an extremal ray of $\overline{\mathrm{NE}}(X)$.

Theorem (Grauert [Gra62])

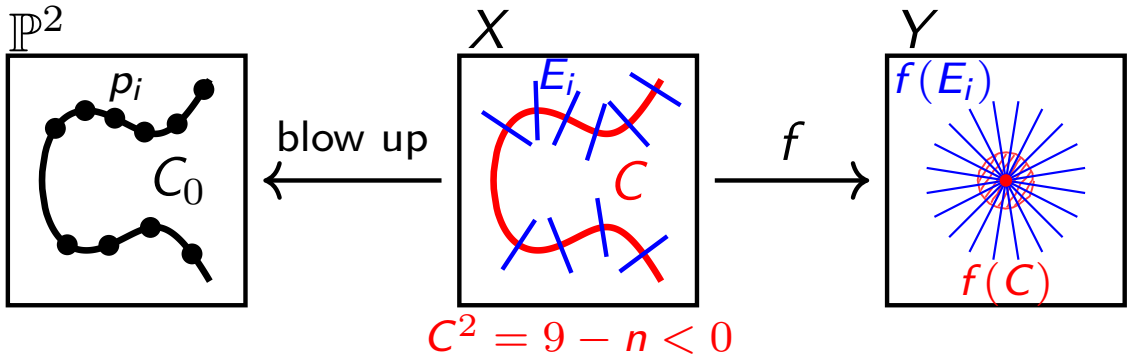
Let $C = \bigcup_i C_i$ be a compact connected curve on a smooth complex surface X. There is a proper bimeromorphic morphism $f : X \to Y$ onto a normal surface contracting exactly C to a point if and only if the matrix $(C_i \cdot C_j)$ is negative definite.



In general Y is only a normal compact complex surface. Is Y projective?

Projectivity of the contraction: two cases

Case 1: C **rational.** If $C \cong \mathbb{P}^1$, then Y is projective (Artin [Art62]).
Case 2: C **elliptic.** Then Y need not be projective.



If Y is projective with H ample, then f^*H is a big line bundle on X, trivial on C. Whether such a line bundle exists depends on C.

Example

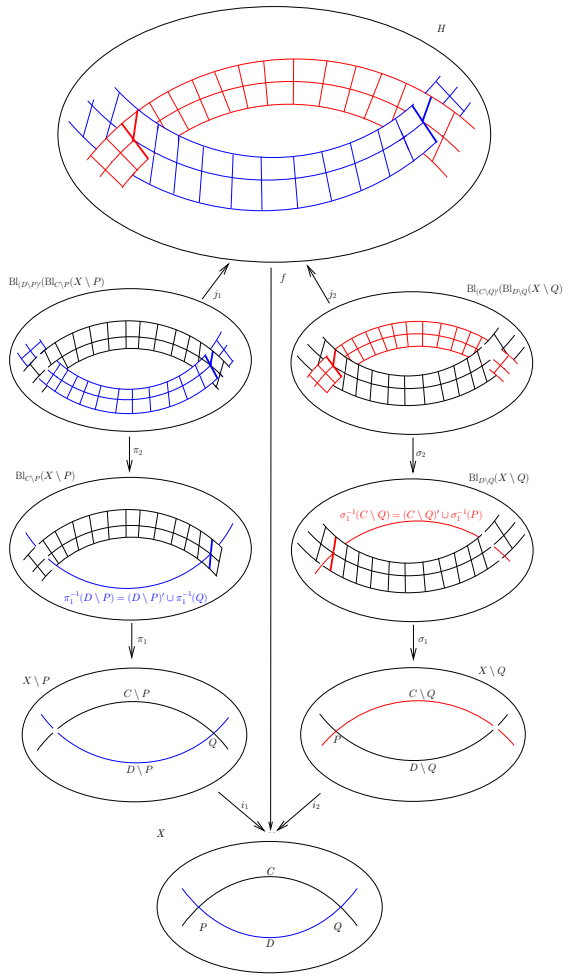
Let X be $\mathbb{P}^2$ blown up at $n \geq 10$ very general points p_i of a smooth cubic C_0, and C its strict transform, so $C^2 = 9 - n < 0$. A line bundle $L = aH - \sum b_i E_i$ is trivial on C only if $a\,H|_C \sim \sum b_i p_i$; very general points satisfy no such relation, so L is trivial and Y is not projective.

Hironaka's example

Let C, D be smooth rational curves in a smooth projective threefold X, meeting transversally at two points P, Q.

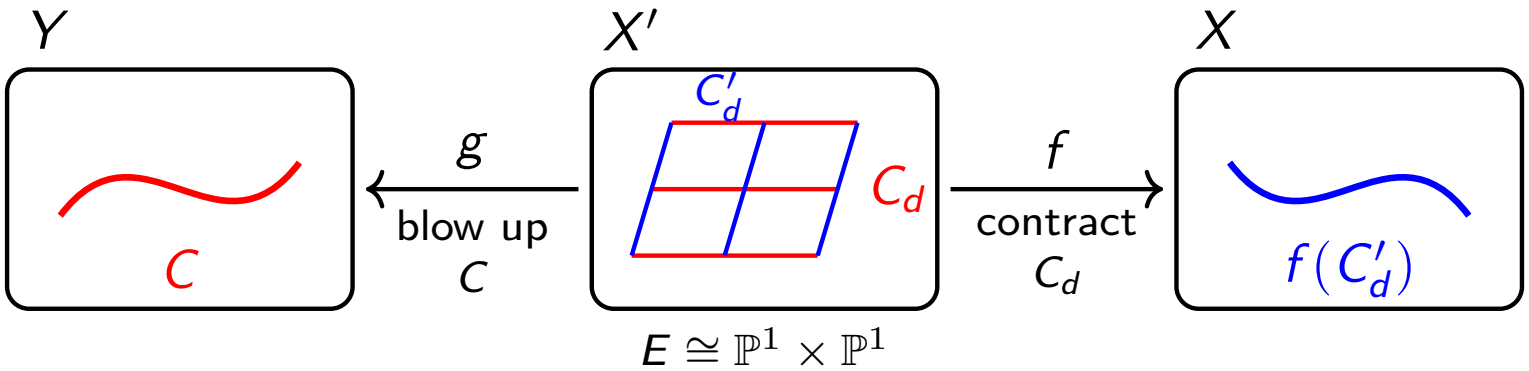
Near P, blow up C and then D; near Q, blow up D and then C. Gluing gives a smooth compact Moishezon threefold $H \to X$, the *Hironaka variety*; in particular H is in Fujiki's class $\mathcal{C}$.

Over P and Q there are rational curves L_P, L_Q with $L_P + L_Q \equiv 0$. Hence H is not Kähler.
Note that $-[L_Q] \equiv [L_P] \in \overline{\mathrm{NA}}(H)$, the cone of positive currents (defined below); this is the obstruction to being Kähler.



A non-Kähler example: flopping a curve on a quintic

Let Y be a general projective quintic threefold and $C \subset Y$ a smooth rational curve of degree d with normal bundle $\mathcal{O}(-1) \oplus \mathcal{O}(-1)$. Blow up C, then contract the exceptional divisor E along the other ruling; the result X is a smooth Moishezon threefold which is not Kähler:



X is not Kähler/projective. For a line $L \subset Y$ disjoint from C, we have

$$C_d \equiv C'_d + d\,L$$

on X'. As f contracts C_d, $-[f(C'_d)] = d\,[f(L)] \in \overline{\mathrm{NA}}(X)$. A Kähler class ω would give $\omega \cdot f(C'_d) > 0$ and $-\omega \cdot f(C'_d) \geq 0$, impossible. So X is not Kähler, hence not projective.

Positivity of classes

Let X be a compact normal variety in Fujiki's class $\mathcal{C}$, and fix a hermitian form ω on X. We work in the *Bott–Chern cohomology* $H^{1,1}_{\mathrm{BC}}(X)$.

- A class α is *Kähler* if it contains a Kähler form;
- A class α is *nef* if for every $\epsilon > 0$ it contains a form $\geq -\epsilon\omega$;
- A class α is *big* if it contains a Kähler current $T \geq \epsilon\omega$;
- A class α is *pseudo-effective* if it contains a closed positive current.

The *generalized Mori cone* $\overline{\mathrm{NA}}(X) \subset H^{1,1}_{\mathrm{BC}}(X)^*$ is the closed cone generated by the classes of closed positive currents of bidimension $(1,1)$.

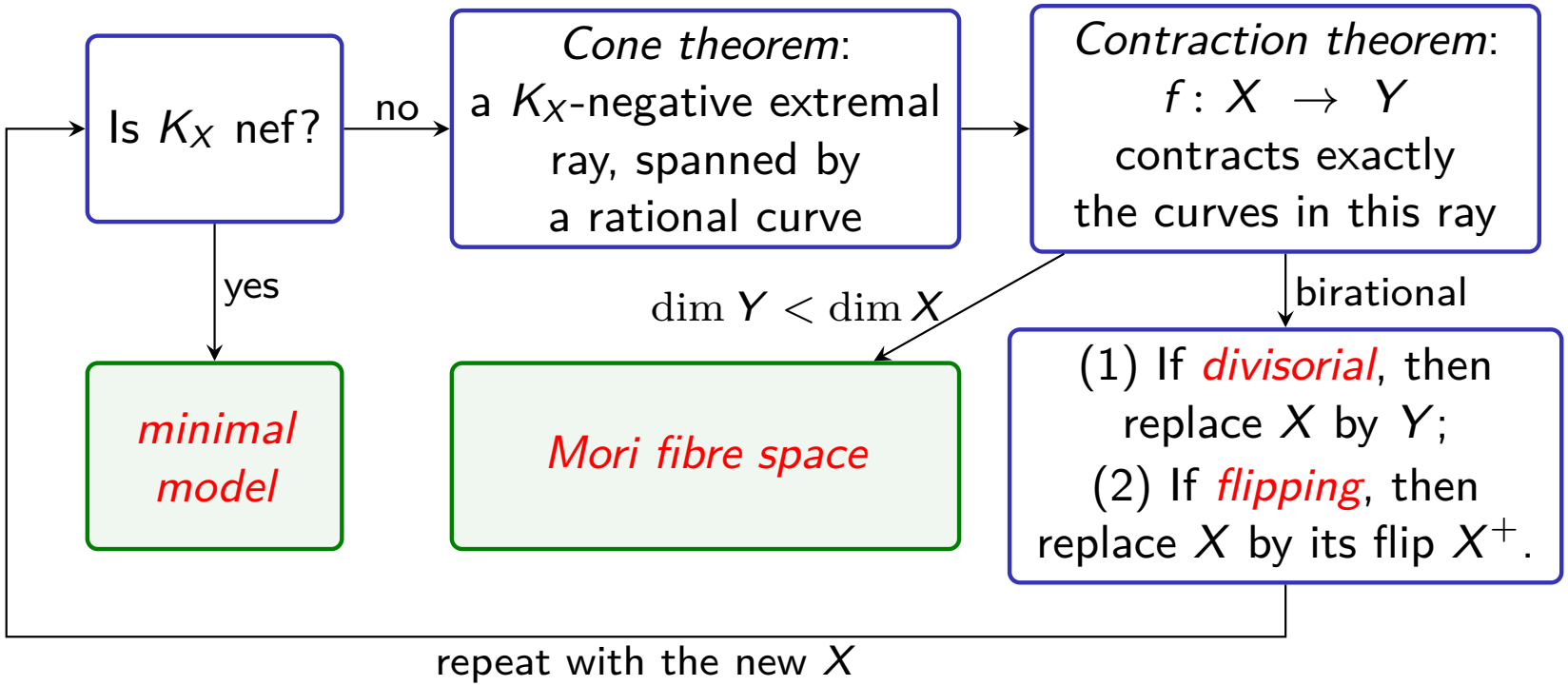
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The minimal model program

Let X be a projective/Kähler variety with klt singularities. The minimal model program goes as follows:



For klt pairs (X, B) with B big, the loop (with scaling) terminates [BCHM10]; see below.

Rational curves and the cone theorem

Theorem (Mori, Kawamata, Kollár, Shokurov)

Let (X, B) be a projective klt pair. Then

$$\overline{\mathrm{NE}}(X) = \overline{\mathrm{NE}}(X)_{K_X+B\geq 0} + \sum_i \mathbb{R}_{\geq 0}[C_i],$$

with C_i rational curves, $0 < -(K_X + B) \cdot C_i \leq 2 \dim X$.

Theorem (Hacon–Păun [HP24])

Let $(X, B + \beta)$ be a compact Kähler generalized klt pair, with X strongly $\mathbb{Q}$-factorial. Then

$$\overline{\mathrm{NA}}(X) = \overline{\mathrm{NA}}(X)_{K_X+B+\beta_X\geq 0} + \sum_i \mathbb{R}_{\geq 0}[\Gamma_i],$$

with Γ_i rational curves, $0 < -(K_X + B + \beta_X) \cdot \Gamma_i \leq 2 \dim X$.

Here a *generalized pair* $(X, B + \beta)$ [BZ16, DHY23] is a boundary divisor B together with a closed b-$(1,1)$-current β which is nef on some higher model.

The contraction theorem

Theorem (Kawamata, Shokurov)

Let (X, B) be a projective klt pair and $R \subset \overline{\mathrm{NE}}(X)$ a $(K_X + B)$-negative extremal ray. Then there is a contraction $f : X \to Y$ onto a normal projective variety which contracts exactly the curves with class in R.

Theorem (Hacon–Xie [HX26])

Let $(X, B + \beta)$ be a compact Kähler generalized klt pair, with X strongly $\mathbb{Q}$-factorial, and $R \subset \overline{\mathrm{NA}}(X)$ a $(K_X + B + \beta_X)$-negative extremal ray. Then the contraction $f : X \to Y$ of R exists and f is projective.

Whether Y is again Kähler is the question we will answer in this talk.

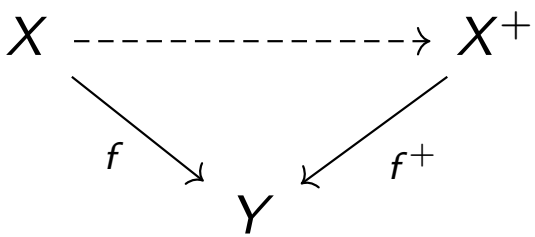
Existence of flips

Theorem (Hacon–McKernan [HM10])

Let (X, B) be a projective $\mathbb{Q}$-factorial klt pair and $f : X \to Y$ a flipping contraction. Then the flip X^+ exists, and X^+ is projective.

Theorem (Das–Hacon [DH25])

Let $(X, B + \beta)$ be a compact Kähler generalized klt pair, with X strongly $\mathbb{Q}$-factorial, and $f : X \to Y$ a flipping contraction. Then the flip $f^+ : X^+ \to Y$ exists, and f^+ is projective.



The Kähler minimal model program

Theorem (Birkar–Cascini–Hacon–McKernan [BCHM10])

Let (X, B) be a projective klt pair with B big. If $K_X + B$ is pseudo-effective, it has a good minimal model; otherwise it has a Mori fibre space.

Conjecture (Kähler MMP [HX26])

Let $(X, B + \beta)$ be a compact Kähler generalized klt pair with $B + \beta_X$ modified big. If $K_X + B + \beta_X$ is pseudo-effective, it has a good minimal model; otherwise it has a Mori fibre space.

Following BCHM, each step needs cone theorem [HP24], contraction theorem [HX26] and existence of flips [DH23, DH25], and:

Question

Is the output of each flip and divisorial contraction again Kähler?

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Main results

Joint work with Christopher D. Hacon and Lingyao Xie.

Throughout, $(X, B + \beta)$ is a generalized klt pair with X compact.

Theorem A (Kähler MMP preserves Kähler condition)

Let X be Kähler and strongly $\mathbb{Q}$-factorial, with $B + \beta_X$ big. Then every flip and every divisorial contraction of X is again Kähler.

Theorem B (A Kählerness criterion)

Let X be in Fujiki's class $\mathcal{C}$ and strongly $\mathbb{Q}$-factorial. Then X is Kähler if and only if there is no rational curve $C \subset X$ with $-[C] \in \overline{\mathrm{NA}}(X)$.

The key tool: a Kähler criterion for adjoint classes

Theorem (Hacon–L.–Xie)

Let $(X, B+\beta)$ be a compact generalized klt pair with X in Fujiki's class $\mathcal{C}$, and $\alpha = [K_X + B + \beta_X]$ nef, big (and NQC). If α is not Kähler, then there is a rational curve $C \subset X$ with $\alpha \cdot C = 0$.

NQC: $\alpha = \sum_i a_i \alpha_i$ with $a_i > 0$ and $\alpha_i \in H^2(X, \mathbb{Q})$ non-negative on $\overline{\mathrm{NA}}(X)$ [HX26, Definition 2.43].

Idea of the proof. Suppose α is not Kähler. We proceed in three steps.

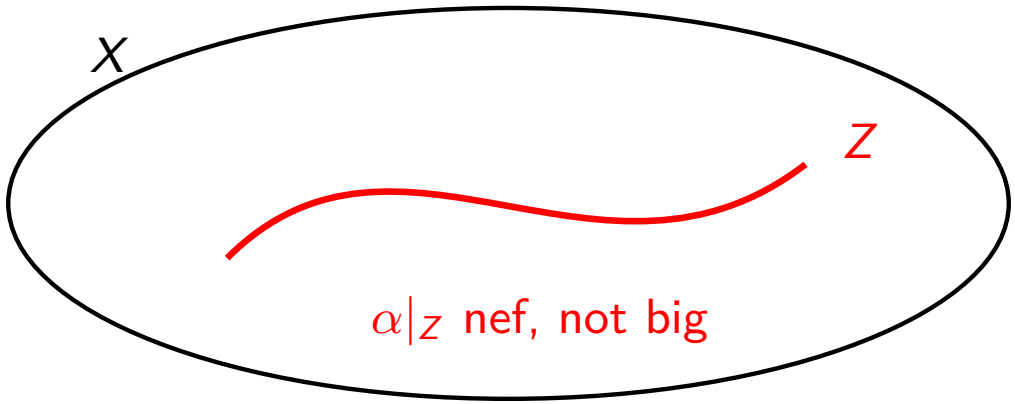
1. α is Kähler away from its null locus [CT15], so there is a component $Z \subset \mathrm{Null}(\alpha)$ on which $\alpha|_Z$ is not big.
2. We show that Z is uniruled. Otherwise K_Z would be pseudo-effective [Ou25], and then $\alpha|_Z$ would be big.
3. On a general fibre of the MRC fibration of Z, α is nef but not big; an MMP there produces an α-trivial rational curve.

Step 1: the null locus

The *null locus* of the nef and big class α is

$\mathrm{Null}(\alpha) = \bigcup\{V \subset X \text{ positive-dimensional subvarieties} : \int_V \alpha^{\dim V} = 0\}$.

If α is not Kähler, then $\mathrm{Null}(\alpha) \neq \emptyset$.



Let Z be a component of $\mathrm{Null}(\alpha)$ of maximal dimension. Then $\alpha|_Z$ is nef but not big, and we look for the rational curve inside Z.

Step 2a: adjunction to a divisor over Z

Z may have higher codimension (not codimension 1), so we cannot do adjunction to Z itself. Instead we replace Z by a divisor S lying over it.

- *Make Z an lc centre.* Take a Kähler current $\eta \in \alpha$ singular exactly along $\mathrm{Null}(\alpha)$ [CT15]; adding a suitable multiple $c\,\eta$ to the boundary makes Z an lc centre, with a unique lc place: a divisor S on a resolution of X. The new adjoint class is $(1+c)\,\alpha$.
- *Adjunction on S.* Restricting this class to the divisor S gives an adjoint class on S, pulled back from a resolution Z' of Z:

$$\begin{array}{ccccc} S & \subset & X' & & \\ {\scriptstyle f}\downarrow & & \downarrow & & \\ Z' & \longrightarrow & Z & \subset & X \end{array} \qquad K_S + \Delta_S + \boldsymbol{\delta}_S \equiv (1+c)\, f^*\alpha_{Z'},$$

where $\alpha_{Z'}$ is the pullback of $\alpha|_Z$ and $\boldsymbol{\delta}_S$ is Kähler.

Step 2b: Z is uniruled

Suppose Z is not uniruled; we show that $\alpha|_Z$ is big, a contradiction. Writing $K_S = K_{S/Z'} + f^*K_{Z'}$ in the adjunction formula, for a small Kähler class $\omega_{Z'}$,

$$(1+c)\, f^*\alpha_{Z'} - f^*\omega_{Z'} \equiv \underbrace{f^*K_{Z'}}_{\text{psef by [Ou25]}} + \underbrace{\left(K_{S/Z'} + \Delta_S + \boldsymbol{\delta}_S - f^*\omega_{Z'}\right)}_{\text{relative part: psef?}}.$$

If the relative part is pseudo-effective, then so is $(1+c)\,\alpha_{Z'} - \omega_{Z'}$, and $\alpha_{Z'}$ is big. The relative part is controlled by:

Theorem (Hacon–Păun [HP24, Theorem 2.2])

Let $g\colon V \to W$ be a projective fibration of compact Kähler varieties and $(V, B+\boldsymbol{\beta})$ generalized lc over general points of W. If

$$K_V + B + \boldsymbol{\beta}_V \equiv g^*\gamma + L \quad \text{with } L \geq 0,$$

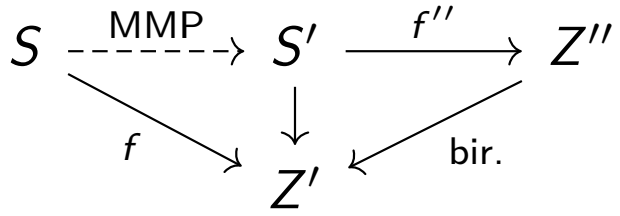
then $K_{V/W} + B + \boldsymbol{\beta}_V$ is pseudo-effective.

Here W is smooth, and L is an effective $\mathbb{R}$-Cartier divisor or a $\mathbb{Q}$-line bundle with sections on the general fibre.

Step 2c: the relative MMP

Hacon–Păun needs a *boundary* whose adjoint class is a *pullback* from the base.

- $K_S + \Delta_S + \boldsymbol{\delta}_S$ is a pullback, but Δ_S may have coefficients ≥ 1 or < 0.
- Truncate the boundary, $\tilde{\Delta}_S = \Delta_S - \Delta_S^{\geq 1} + \Delta_S^{<0}$, and run the MMP of $K_S + \tilde{\Delta}_S + \boldsymbol{\delta}_S$ over Z' to make it a pullback again:



$$K_{S'} + \tilde{\Delta}_{S'} + \boldsymbol{\delta}_{S'} = (f'')^* \lambda'',$$

where S' is a good minimal model, f'' its lc model and $\lambda'' \in H^{1,1}_{\mathrm{BC}}(Z'')$ is Kähler over Z'.

Hacon–Păun applies on S', so the relative part is pseudo-effective and $\alpha_{Z'}$ is big, a contradiction. So Z is uniruled.

Step 3: finding the rational curve

Let $\varphi\colon Z' \dashrightarrow W$ be the MRC fibration, with general positive-dimensional fibre F; it is projective.

$$\begin{array}{ccccc} Z' & \overset{\varphi}{\dashrightarrow} & W & & \text{not uniruled} \\ \uparrow & & & & \\ F & \overset{\text{MMP}}{\dashrightarrow} & F_m & \xrightarrow{\ g\ } & T \end{array}$$

rationally connected

- We first show that $\alpha_F = \alpha_{Z'}|_F$ is nef, not big. Otherwise, as W is not uniruled, the argument of the previous step makes $\alpha_{Z'}$ big.
- For $t \gg 0$, the $(K_F + t\,\alpha_F + \omega_F)$-MMP ends with a Mori fibre space $g\colon F_m \to T$.
- By the NQC condition, the MMP is α_F-trivial and F_m is covered by α_{F_m}-trivial rational curves, therefore F is covered by α_F-trivial rational curves.

□

Theorem A: the Kähler MMP stays Kähler

Theorem A (Hacon–L.–Xie)

Let $(X, B + \beta)$ be a generalized klt pair, with X compact Kähler and strongly $\mathbb{Q}$-factorial and $B + \beta_X$ big. If $f : X \to Z$ is a flipping or divisorial contraction, then Z is Kähler, and so is the flip X^+.

Theorem A is a corollary of the Kähler criterion, applied on Z.

1. f contracts an extremal ray R, cut out by a nef class $\alpha = [K_X + B + \beta_X + \omega]$, ω Kähler: $\alpha \cdot C = 0 \iff [C] \in R$. It is big, since f is birational.
2. Descend α to Z: $\alpha = f^*\alpha_Z$, where $\alpha_Z = [K_Z + B_Z + \beta_Z + f_*\omega]$ is again an adjoint class on Z, nef and big.
3. A curve $C \subset Z$ is the image of a curve $C' \subset X$ not in R, so $\alpha_Z \cdot C = \alpha \cdot C' > 0$. With no α_Z-trivial curve, α_Z is Kähler.
4. *The flip.* $X^+ \to Z$ is projective and Z is Kähler, so X^+ is Kähler. □

Theorem B: when is a Fujiki variety Kähler?

Theorem B (Hacon–L.–Xie)

Let $(X, B+\beta)$ be a generalized klt pair with X compact in Fujiki's class $\mathcal{C}$. If X is not Kähler, then either

1. *X contains a rational curve C with $-[C] \in \overline{\mathrm{NA}}(X)$, or*
2. *there is a small $\mathbb{Q}$-factorial Kähler modification $X^{\mathrm{qf}} \to X$.*

Corollary

If moreover X is strongly $\mathbb{Q}$-factorial, then X is Kähler if and only if there is no rational curve C with $-[C] \in \overline{\mathrm{NA}}(X)$.

Theorem B follows from Theorem A: an MMP over X from a Kähler resolution stays Kähler, and ends in (2) or produces the curve in (1).

Thank you for your attention!

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